



ADARSH MURALI

Data | Analyst | Business Intelligence



Experience : Analytics|Data Engineering|Data Warehousing| Backend-development| Big-Data|Business Intelligence| Visualization & Reporting

Interests : Real Time Streaming Analytics | Investment Banking

Ph : 9611533665 Email : adarsh.5.88@gmail.com

Experience

15+

Total Years

13+

Data Engineer

5+

Python Development

13+

Business Intelligence

8+

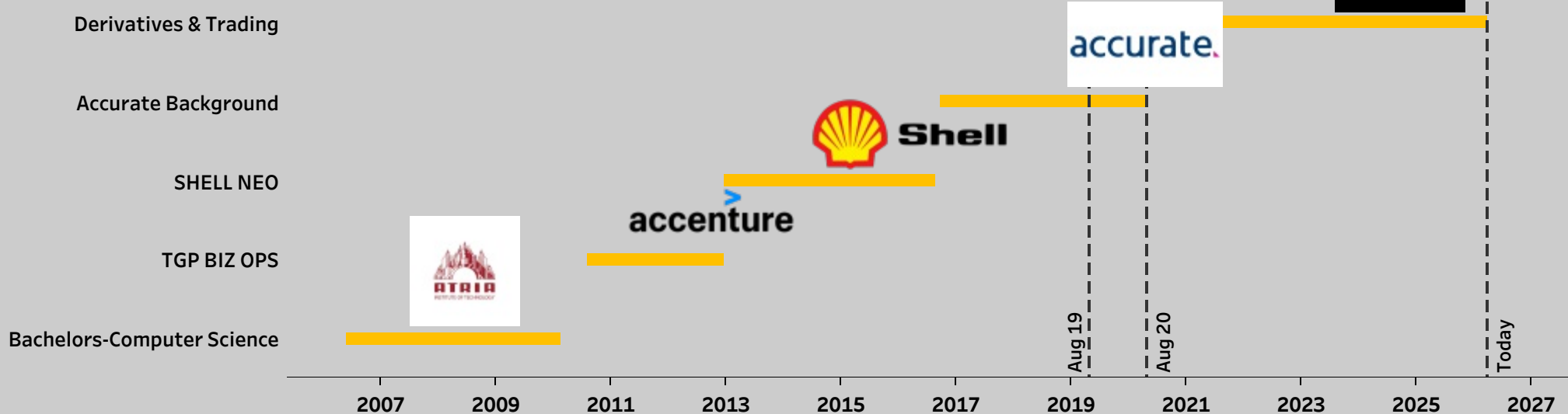
Finance

9+

Agile Projects



TIMELINE



FINSIGHT AI | Hedge Fund Portfolio Analyzer

PERSONAL BUILD . 2026

DOMAIN

Hedge Fund Portfolio Analytics · Capital Markets · Investment Banking

Target Users

Hedge Funds, Institutional Investors, Portfolio Managers

Overview

An agentic AI-powered financial research platform for hedge funds and institutional investors, combining real-time streaming analytics with GPT-4o-driven portfolio intelligence. Enables natural language querying of live portfolio data, risk metrics, and market events — replacing manual analyst work with autonomous tool-calling AI.

Key Highlights

Built an **agentic AI chat system** using **GPT-4o function calling** (not LangChain — native OpenAI tool loop), where the model autonomously decides which of 5 tools to invoke: portfolio analysis, position history, RAG market search, market events lookup, and risk analysis

Built a **FastAPI backend** with 18+ REST endpoints and WebSocket support; SQLAlchemy ORM over **SQL Server 2022** (Docker); covers portfolio CRUD, analytics, AI insights, alerts, and risk

Implemented a **real-time streaming pipeline**: **Finnhub WebSocket** → **Apache Kafka** (market.trades, market.news topics) → **Apache Flink** (Python) → **ChromaDB** — detecting live volume/price spikes across 50 tickers with sub-5-minute latency; validated 9,963+ trade messages and 128 volatility events on first live run

Built a **RAG system using ChromaDB** with 24,000+ documents across 10 collections (OHLV, macro indicators, SEC filings, live market news, earnings, analyst research, technical indicators, dividends, splits, volatility events); semantic search feeds grounded context into every GPT-4o response

Implemented **Advanced Risk Analytics**: Historical VaR (95%/99%, 1-day/10-day), Parametric VaR, 6 stress test scenarios (2008 Crisis, COVID Crash, Rate Shock, Dot-com Bust, Oil Shock, Stagflation), and factor exposure regression vs SPY/QQQ/VTV/MTUM

Built a **threshold-based alert engine** that fires on VaR and stress test breaches with deduplication; automated daily risk job via **AWS EventBridge Scheduler** → **SSM** → **EC2** (Mon–Fri 17:30 ET)

Built a **Next.js frontend** with Bloomberg Terminal-style UI (black/orange, monospace); live portfolio value flashing green/red via WebSocket; Kafka news surfaces as amber toast alerts; AI chat with streaming responses and orange tool-call chips showing which tools fired

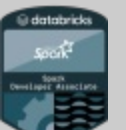
Designed **SQL Server schema** with 8 tables (Customers, Portfolios, Positions, Securities, Transactions, Market Events, Portfolio Performance, Position Changes Log); generated synthetic institutional dataset: 1,500+ portfolio records, 500+ securities, 12 months of history

Data sources integrated: **OpenAI** (GPT-4o), **Finnhub** (live trades/news), **FRED** (macroeconomic data), **Alpha Vantage** (market data), **yfinance** (historical OHLV)

Technology: Python, FastAPI, Apache Flink, Apache Kafka, SQL Server 2022, ChromaDB, OpenAI GPT-4o, Next.js, Docker, AWS EC2, AWS EventBridge, SSM, yfinance, FRED API, Finnhub API, SQLAlchemy, Tailwind CSS



CERTIFICATIONS



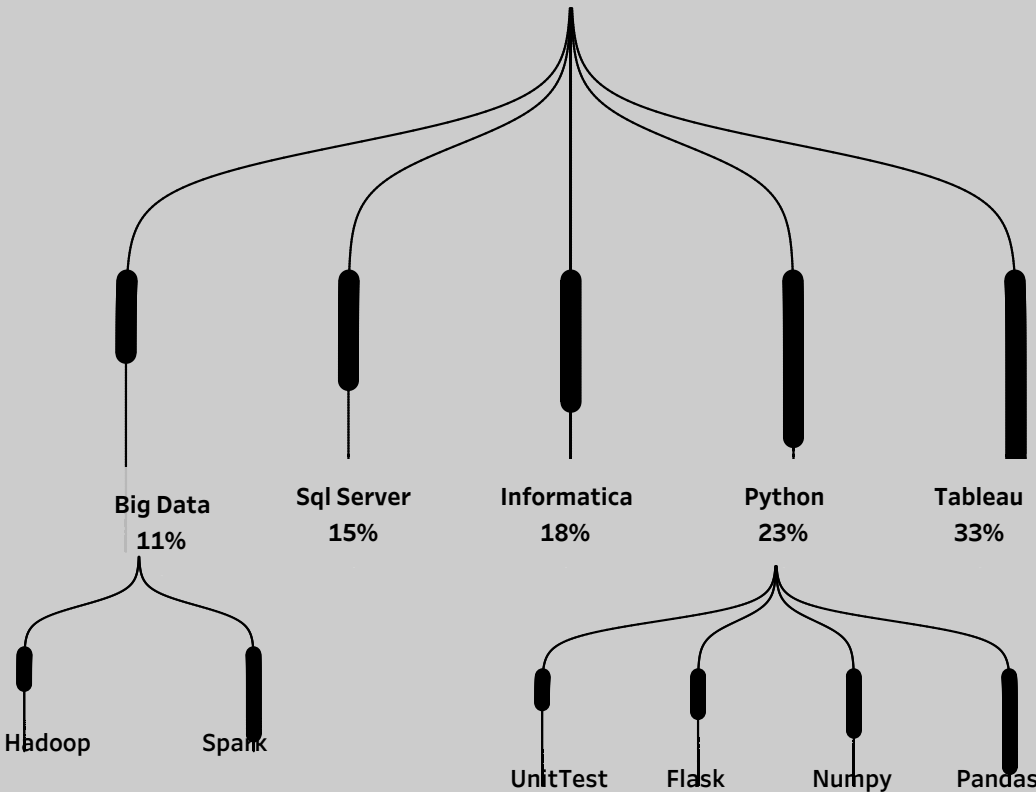
SKILLS

Tableau	●	●	●	●	○
SQL Server	●	●	●	●	○
Microsoft BI	●	●	●	●	○
Python	●	●	●	○	○
RedShift	●	●	●	○	○
Spark	●	●	●	○	○
Informatica	●	●	●	○	○
AWS	●	●	●	○	○
Github Copilot	●	●	●	○	○
Google Cloud	●	●	○	○	○
Azure	●	●	○	○	○
Docker	●	●	○	○	○

TOOLS USED DAILY



INDUSTRY



WELLSFARGO

Seasoned Data Engineer at Wells Fargo's Capital Markets Division, specializing in Dodd-Frank Act compliance and Section 165(d) submissions for living will requirements-resolution planning. Involved in timely regulatory reporting for Federal submission.

I have used Informatica for Ingesting data into Sql Server and used python as backend programming language and for api creation & written code coverage, used tableau for regulatory fed submission, used hadoop/spark for storing the data for auditing and reporting use cases. Github Copilot is used for development work.

I have participated in Technology Spotlight events and presented Sessions to entire wells Fargo india on Spark Streaming. I have utilized Microsoft Azure - Devops for Hackathon 2023. I have worked on Google Cloud Platform - Vertex AI - for Hackathon 2024. I have worked on OpenShiftContainerPlatform for Hacathon 2025. I have completed the derivatives training as part of Markets College Training program - a Capital Markets Initiative

Technology : Informatica, Tableau, Python, Sql Server, Hadoop, Spark

ENCORA LABS(CLIENT : ACCURATE)

Key member of Accurate's IT Business Intelligence and Analytics team, specializing in data cleansing, transformation, and the application of business logic for analytics purposes. Played a pivotal role in ensuring data accuracy and relevance for major clients such as Amazon, Uber, and Starbucks. Proficient in leveraging data-driven insights to drive business decisions and enhance operational efficiency. Demonstrated expertise in project lifecycle management, from conceptualization to deployment and post-launch support I have used AWS-Redshift along with Tableau to fulfill real time reporting needs.

Technology : Microsoft-BI(SSIS, SSRS), Sql Server, Tableau, MongoDB, AWS-Redshift, AWS-DMS, Dimensional Modelling

ACCENTURE(CLIENT : SHELL)

Contributed to Shell's IT department as a data specialist, focusing on data cleaning, preparation, and extraction of actionable insights. Spearheaded project initiation, including requirements gathering, feasibility analysis, and stakeholder alignment to establish project scope and objectives. Managed the transition to production, collaborating with the production support team to facilitate a smooth handover and ensure operational readiness. Provided ongoing support during the post-launch phaseApplied expertise in SSIS, SQL Server, and Oracle to manage data related to crucial aspects of the energy sector such as directional survey and facility monitoring. Additionally, gained exposure to SAP HANA and SAP BODS for enhanced data processing capabilities.

Technology : SSIS, Sql Server, Oracle, SAP BODS/HANA

ACCENTURE

Played a vital role in an internal project at Accenture, focusing on financial reporting and risk analysis for top 100 clients. Led data engineering efforts using SSIS and SQL Server for ETL processes. Additionally, contributed to a certification initiative by managing delivery lead and solution architect programs, enhancing firm-wide operational efficiency

Technology : C#, SSIS, Sql Server, MS Access